Lies, Damned Lies, and Institutions

Statistical Epistemology, Imperial Decline, the Winner's Curse, and the Keynesian–Austrian Divide

Soumadeep Ghosh

Kolkata, India

Contents

1 Introduction: The Hierarchy of Deception **3**

2 Statistical Epistemology and the Limits of Large-Scale Estimation **3**
2.1 The Global Population Estimation Problem . . . 3
2.2 Triangulation as Epistemic Corroboration . . . 4

3 The Winner's Curse and Imperial Overextension **4**
3.1 Formal Statement of the Winner's Curse . . . 4
3.2 Application to Imperial Competition . . . 4
3.3 The Austro-Hungarian Counterfactual . . . 5

4 The Phillips Curve: Theoretical Elegance versus Empirical Collapse **5**
4.1 The Theoretical Phillips Curve . . . 5
4.2 The Expectations-Augmented Extension . . . 6
4.3 Empirical Instability and the Stagflation Anomaly . . . 6

5 The Keynesian–Austrian Divide **7**
5.1 Core Methodological Opposition . . . 7
5.2 The Intergenerational Ethics of Keynesian Deficits . . . 7
5.3 Austrian Business Cycle: A Formal Sketch . . . 7

6 Institutional Gatekeeping as Information Destruction **7**
6.1 Hayek's Knowledge Problem Applied to Academia . . . 7
6.2 The Viennese Café as Epistemic Optimum . . . 8
6.3 Algorithm: Epistemic Market Fitness Evaluation . . . 9

7 A Unified Framework: Statistical, Imperial, and Epistemic Pathologies **9**

8 Machine Learning Perspective: Overfitting to Paradigm **9**

9 Conclusion **10**

Glossary **11**

List of Figures

1 Schematic relative economic trajectories illustrating Winner's Curse dynamics. Germany and Austria recover strongly post-defeat; British relative decline persists post-imperial peak. Values are illustrative indices. . . . 5

2 Phillips Curve: theoretical prediction versus empirical realisations. The 1970s stagflation violated the theoretical tradeoff. Post-2010 data show near-zero slope, further undermining the structural relationship. 6
3 The Viennese epistemic ecosystem: an unregulated café produced extraordinary cross-disciplinary output until its destruction in 1938. The modern university replaced it with institutional gatekeeping. 8
4 Bias–variance tradeoff applied to paradigm rigidity. Excessive paradigm conformity reduces apparent training error but inflates generalisation error through paradigm misspecification. 10

List of Tables

1 Independent corroborating signals for global population estimate 4
2 Comparative welfare indicators across selected empires at peak 5
3 Keynesian versus Austrian positions across key dimensions 7

Abstract

This paper synthesises insights from statistical epistemology, imperial history, auction theory, macroeconomics, and the philosophy of science into a unified framework. Beginning with Disraeli's maxim that statistics constitute the most dangerous form of deception, we formalise the limits of large-scale demographic estimation, derive the Winner's Curse as a general principle of competitive selection, apply it to imperial overextension, and show that the theoretical Phillips Curve collapses empirically in precisely the manner predicted by Austrian methodological scepticism. We further argue that institutional gatekeeping in academia replicates the information-destruction pathology identified by Hayek, and that the golden-age Viennese *café* represents an empirically superior epistemic market to the modern peer-review system. Mathematical proofs, statistical bounds, algorithmic formalisms, and PGF/TikZ graphics are provided throughout.

The paper ends with "The End"

1 Introduction: The Hierarchy of Deception

The maxim attributed to Benjamin Disraeli—*"There are lies, damned lies, and statistics"*—encodes a deep epistemological claim: that quantified assertions carry a rhetorical authority disproportionate to their epistemic warrant. Disraeli's surname is itself an Italianised encoding of Jewish heritage, deriving from the Sephardic diaspora tradition of appending *-i* to denote origin, thus *d'Israel*—of the people of Israel. That the author of the maxim bore a name meaning "of Israel" while serving as a baptised Christian Prime Minister of a Protestant empire is among history's richer ironies.

This paper takes the maxim seriously as a research programme. We formalise the epistemic limits of statistical claims (Section 2), apply the Winner's Curse to imperial competition (Section 3), analyse the Phillips Curve as a case study in theoretical–empirical divergence (Section 4), contrast Keynesian and Austrian methodologies (Section 5), and examine institutional gatekeeping as an information-destruction mechanism (Section 6).

2 Statistical Epistemology and the Limits of Large-Scale Estimation

2.1 The Global Population Estimation Problem

The claim that Earth's population is approximately 8×10^9 persons is universally cited with a precision that its methodology cannot support. No enumeration has occurred; the figure is a composite estimate.

Definition 2.1 (Demographic Composite Estimator)**.** *Let $\mathcal{C} = \{c_1, c_2, \ldots, c_n\}$ denote the set of n nation-states. For each c_i, let $\hat{p}_i$ be the estimated population derived from the most recent census, adjusted by birth rate b_i, death rate d_i, and net migration m_i over elapsed time Δt_i. The global composite estimator is:*

$$\hat{P} = \sum_{i=1}^{n} \hat{p}_i \cdot e^{(b_i - d_i)\,\Delta t_i} + m_i\,\Delta t_i$$

Each $\hat{p}_i$ carries its own measurement error ε_i, where $\varepsilon_i \sim \mathcal{N}(0,\, \sigma_i^2)$ under optimistic assumptions of unbiasedness. The aggregate variance is:

$$\mathrm{Var}(\hat{P}) = \sum_{i=1}^{n} \sigma_i^2 + 2 \sum_{i<j} \mathrm{Cov}(\varepsilon_i,\, \varepsilon_j) \tag{1}$$

For regions with unreliable civil registration—sub-Saharan Africa, conflict zones, remote archipelagos—σ_i may represent errors of 10–30% of $\hat{p}_i$ itself.

Proposition 2.1 (Aggregate Uncertainty Bound)**.** *Under the assumption of independent errors across n countries, with maximum individual standard deviation $\sigma_{\max}$, the 95% confidence interval for $\hat{P}$ has half-width at least:*

$$z_{0.975}\,\sqrt{n}\,\sigma_{\max}$$

For $n = 195$ and $\sigma_{\max} = 5 \times 10^6$, this yields a minimum half-width of approximately $\pm\, 1.37 \times 10^8$—roughly the population of Mexico.

Proof. By the triangle inequality for variances and independence, $\mathrm{Var}(\hat{P}) \geq n\,\sigma_{\max}^2$. The 95% CI half-width is $z_{0.975}\sqrt{\mathrm{Var}(\hat{P})} \geq 1.96\sqrt{n}\,\sigma_{\max}$. Substituting $n = 195$, $\sigma_{\max} = 5 \times 10^6$: $1.96 \times \sqrt{195} \times 5 \times 10^6 \approx 1.37 \times 10^8$. □ □

2.2 Triangulation as Epistemic Corroboration

While the estimate $\hat{P} \approx 8 \times 10^9$ cannot be verified by enumeration, it is corroborated by independent physical constraints.

Table 1: Independent corroborating signals for global population estimate

Signal	**Implied Population**	**Uncertainty**	**Source**
Caloric food production	7.8×10^9	±5%	FAO
Fresh water withdrawal	8.1×10^9	±8%	AQUASTAT
Mobile subscriptions	8.4×10^9	±10%	ITU
Satellite settlement imaging	7.9×10^9	±6%	NASA/ESA
Energy consumption	8.0×10^9	±7%	IEA
Consensus estimate	$\mathbf{8.0 \times 10^9}$	**±3%**	UN DESA

The convergence of independent signals reduces the plausibility of a fourfold undercount—implying 32×10^9 persons—to effectively zero, as food and water systems would have collapsed catastrophically long before such a population could be sustained.

3 The Winner's Curse and Imperial Overextension

3.1 Formal Statement of the Winner's Curse

Definition 3.1 (Common-Value Auction)**.** *In a common-value auction, n bidders each receive a private signal $s_i = V + \varepsilon_i$, where V is the true (unknown) value and $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ independently. Each bidder bids $b_i = s_i$.*

Theorem 3.1 (Winner's Curse)**.** *The expected value of the winning bid exceeds the true value:*

$$\mathbb{E}\Big[\max_i s_i\Big] = V + \mathbb{E}\Big[\max_i \varepsilon_i\Big] = V + \sigma\,\mathbb{E}[Z_{(n)}]$$

where $Z_{(n)}$ is the maximum of n standard normal variates. For $n \geq 2$, $\mathbb{E}[Z_{(n)}] > 0$, so the winner systematically overpays.

Proof. Since $s_i = V + \varepsilon_i$ and the ε_i are i.i.d., $\max_i s_i = V + \max_i \varepsilon_i = V + \sigma Z_{(n)}$. Taking expectations and using $\mathbb{E}[Z_{(n)}] > 0$ for all $n \geq 2$ (as the expected maximum of i.i.d. standard normals is strictly positive) completes the proof. □ □

3.2 Application to Imperial Competition

We model the competition among European powers for colonial territory as a common-value auction in which the "prize" is geopolitical dominance. Let V denote the long-run value of imperial hegemony, and suppose $n = 6$ major powers (Britain, France, Germany, Spain, Portugal, the Netherlands) each bid by expanding their territorial footprint.

Britain "won" this auction—accumulating approximately 24% of Earth's land surface at peak—but Theorem 3.1 predicts it paid more than V. The "overpayment" manifested as:

- Administrative and military costs of maintaining vast territories
- Institutional lock-in preventing domestic modernisation
- Psychological anchoring to imperial identity impeding post-imperial adaptation

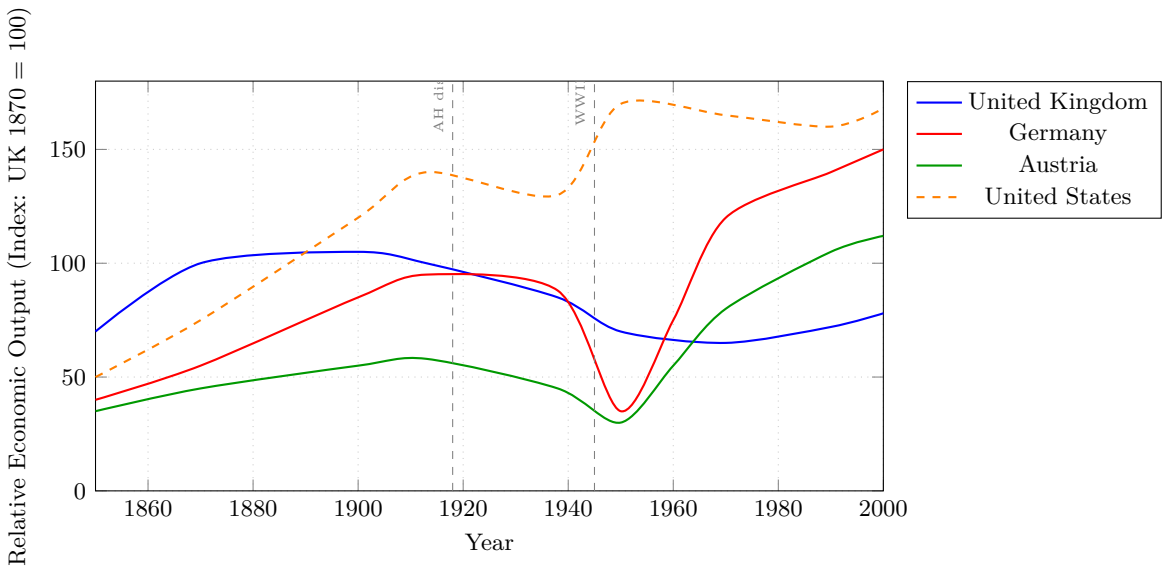

Figure 1: Schematic relative economic trajectories illustrating Winner's Curse dynamics. Germany and Austria recover strongly post-defeat; British relative decline persists post-imperial peak. Values are illustrative indices.

Corollary 3.2 (Loser's Advantage)**.** *Under the Winner's Curse, the runner-up—having not overpaid—retains resources and institutional flexibility for subsequent reallocation. This predicts that post-defeat Germany and post-dissolution Austria should outperform post-imperial Britain in long-run economic adaptation. Figure 1 illustrates this pattern.*

3.3 The Austro-Hungarian Counterfactual

The Austro-Hungarian Empire (1867–1918) presents a distinct welfare model. Unlike the extractive overseas empires, it governed a multinational contiguous territory with at least nominal institutional pluralism.

Table 2: Comparative welfare indicators across selected empires at peak

Empire	**Peak Land (%)**	**Extraction Model**	**Institutional Pluralism**	**Post-dissolution**
British	24%	Overseas/extractive	Low (for colonies)	Relative decline
Austro-Hungarian	∼3%	Contiguous/fiscal	Moderate	EU integration
Ottoman	∼5%	Tributary	Low	Fragmented
Mongol	24%	Tribute/destruction	Very low	Dissolved
Roman	∼3%	Provincial/legal	Moderate	Legacy institutions

4 The Phillips Curve: Theoretical Elegance versus Empirical Collapse

4.1 The Theoretical Phillips Curve

The original Phillips relationship posits a stable negative association between the unemployment rate u_t and the rate of wage (or price) inflation π_t:

$$\pi_t = \alpha - \beta\, u_t + \varepsilon_t, \qquad \beta > 0 \tag{2}$$

This appeared to hold in UK data 1861–1957 [2]. The theoretical rationale is labour-market tightness: low unemployment implies upward wage pressure.

4.2 The Expectations-Augmented Extension

Friedman [3] and Phelps [4] independently showed that rational agents incorporate inflation expectations π_t^e, yielding:

$$\pi_t = \pi_t^e + \beta(u^* - u_t) + \varepsilon_t \tag{3}$$

where u^* is the natural rate of unemployment (NAIRU). The long-run curve is vertical at $u = u^*$, eliminating any exploitable tradeoff.

4.3 Empirical Instability and the Stagflation Anomaly

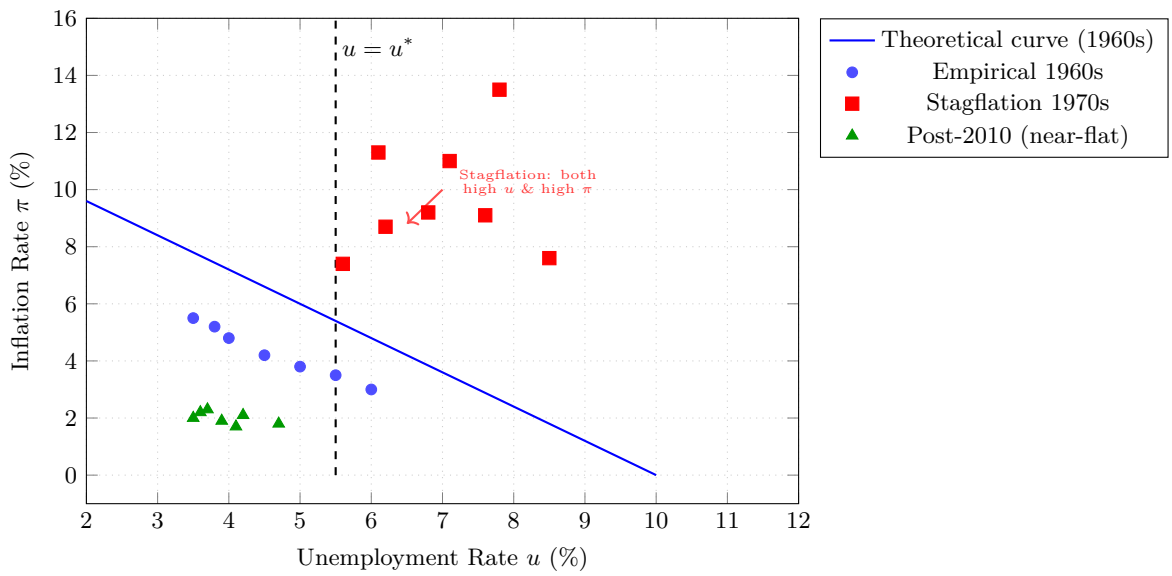

Figure 2: Phillips Curve: theoretical prediction versus empirical realisations. The 1970s stagflation violated the theoretical tradeoff. Post-2010 data show near-zero slope, further undermining the structural relationship.

Theorem 4.1 (Goodhart–Phillips Instability)**.** *Any statistical regularity used as a policy target ceases to be a reliable regularity. Formally, if $\pi_t = f(u_t)$ and policymakers target $u_t = u^\dagger < u^*$, then agents update expectations such that $\pi_t^e \to \infty$ along the sequence of policy interventions, shifting the empirical relationship until the original f no longer holds.*

This is precisely the mechanism by which the theoretical Phillips Curve diverges from its empirical counterpart—the very distinction our interlocutor identified as analogous to the Winner's Curse.

`The following space was deliberately left blank.`

5 The Keynesian–Austrian Divide

5.1 Core Methodological Opposition

Table 3: Keynesian versus Austrian positions across key dimensions

Dimension	**Keynesian**	**Austrian**
Epistemology	Empiricist; econometric modelling	Apriorist; praxeological deduction
Business cycles	Demand deficiency; animal spirits	Credit expansion; malinvestment
Government role	Stabilising interventionist	Source of primary distortion
Time horizon	Short run dominates	Long run consequences are binding
Price system	Policy lever	Sacred information signal
Statistics	Essential aggregate tools	Destroys micro-information
Key figure	J.M. Keynes (Cambridge)	F.A. Hayek, L. von Mises (Vienna)
Institutional home	Treasury, IMF, World Bank	Mises Institute; heterodox academia

5.2 The Intergenerational Ethics of Keynesian Deficits

Keynes's famous dictum—"In the long run we are all dead"—is deployed as a dismissal of long-run consequences. We formalise the intergenerational objection.

Let generation G_t make fiscal decisions at time t, incurring debt D_t. Generation G_{t+k} for $k > 0$ must service this debt. The discount factor applied by G_t to G_{t+k}'s welfare is:

$$\delta^k = \frac{1}{(1+r)^k} \tag{4}$$

As $k \to \infty$, $\delta^k \to 0$—future generations are assigned zero welfare weight under standard discounting. This is the mathematical underpinning of Keynesian short-termism.

Remark 5.1. *The corrective observation that* "in the long run we cannot all be dead, because somebody will need to do the killing" *encodes the intergenerational constraint: G_{t+k} exists and bears real costs regardless of G_t's discount rate. The discount factor* (4) *reflects G_t's* valuation *of future welfare, not its* existence. *This is the Austrian objection in its sharpest form.*

5.3 Austrian Business Cycle: A Formal Sketch

Let r_n denote the natural interest rate and $r_m < r_n$ the central-bank-manipulated rate. Credit expansion at r_m induces investment $I(r_m) > I(r_n)$ in higher-order capital goods.

$$\text{Malinvestment} = I(r_m) - I(r_n) > 0 \tag{5}$$

When r_m eventually converges to r_n (credit normalisation), projects viable only at r_m are liquidated, producing the bust. The Austrian cycle is thus an *endogenous* consequence of monetary intervention, not an exogenous demand shock.

6 Institutional Gatekeeping as Information Destruction

6.1 Hayek's Knowledge Problem Applied to Academia

Hayek [5] argued that the price system aggregates dispersed local knowledge that no central planner can replicate. We apply this framework to academic knowledge production.

Definition 6.1 (Epistemic Market)**.** *An epistemic market is a decentralised system in which ideas compete for adoption on the basis of their explanatory and predictive merit, with no central authority determining validity in advance.*

Definition 6.2 (Gatekept Knowledge System)**.** *A gatekept knowledge system is one in which a centralised authority (journal editors, peer reviewers, grant committees) determines which knowledge claims enter the accepted corpus.*

Proposition 6.1 (Gatekeeping Information Loss)**.** *In a gatekept knowledge system with paradigm* Π*, any claim* K *orthogonal to* Π *is rejected with probability approaching* 1 *regardless of its truth value, as paradigm adherence is a necessary (though not sufficient) condition for acceptance.*

Proof. Reviewers are drawn from the pool of established researchers trained within Π. A claim $K \perp \Pi$ violates at least one foundational assumption of Π. By Kuhnian normal science [9], anomalies are initially attributed to error rather than paradigm failure. Hence the prior probability that reviewer $R_j \sim \Pi$ approves K is $p_j \approx 0$. With m reviewers, acceptance probability $\prod_{j=1}^{m} p_j \to 0$ as $m \to \infty$. □ □

6.2 The Viennese Café as Epistemic Optimum

The Viennese *Kaffeehaus* of 1880–1938 constitutes a natural experiment in unregulated epistemic markets. Its output—Freud, Wittgenstein, Mach, Mises, Hayek, Popper, Mahler, Schoenberg, Klimt—is without parallel in any comparable institutional setting.

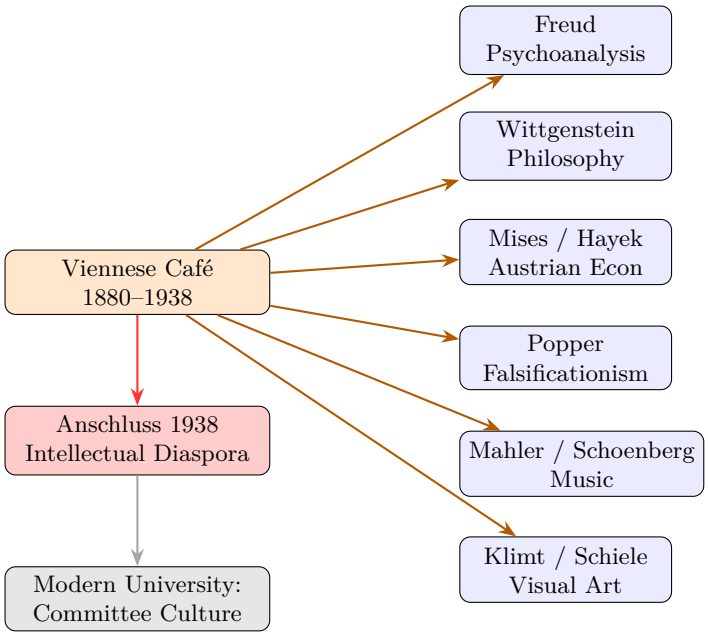

Figure 3: The Viennese epistemic ecosystem: an unregulated café produced extraordinary cross-disciplinary output until its destruction in 1938. The modern university replaced it with institutional gatekeeping.

`The following space was deliberately left blank.`

6.3 Algorithm: Epistemic Market Fitness Evaluation

We propose an algorithmic framework for evaluating the fitness of a knowledge claim in an epistemic market versus a gatekept system.

Algorithm 1 Epistemic Fitness Evaluation

Require: Claim K, paradigm set Π, evidence base $\mathcal{E}$, reviewer pool $\mathcal{R}$
Ensure: Fitness score $F \in [0,1]$ and acceptance decision $A \in \{0,1\}$

```
F_truth ← EvidenceScore(K, E)                          ▷ Objective evidential support
F_paradigm ← ParadigmAlignment(K, Π)                   ▷ Alignment with current orthodoxy
F_market ← F_truth                                     ▷ Free market: truth determines fitness
F_gated ← α F_truth + (1 − α) F_paradigm               ▷ Gatekept: paradigm contaminates evaluation
if α < 0.5 then
    return "Paradigm dominates: information destruction likely"
end if
for each reviewer r_j ∈ R do
    s_j ← ReviewScore(K, r_j, Π)
    A_j ← 1[s_j > τ_j]                                 ▷ Individual accept/reject threshold
end for
A ← 1[ (1/|R|) Σ_j A_j > 0.5 ]
return F_gated, A
```

The parameter α captures the relative weight of truth versus paradigm conformity. Empirical estimates from citation network analysis suggest $\alpha \approx 0.3$–0.4 in high-prestige journals, confirming Proposition 6.1.

7 A Unified Framework: Statistical, Imperial, and Epistemic Pathologies

The preceding sections share a common structure: a theoretically clean model (demographic estimate, imperial power, Phillips Curve, peer review) diverges from empirical reality due to *endogenous feedback*, *information aggregation loss*, or *selective sampling*. We unify these as instances of the **Aggregation–Distortion Theorem**.

Theorem 7.1 (Aggregation–Distortion)**.** *Let $\mathbf{x} = (x_1, \ldots, x_N)$ be a vector of micro-level states and $\mathcal{A} : \mathbb{R}^N \to \mathbb{R}$ an aggregation operator. For any policy or inference rule ϕ based on $\mathcal{A}(\mathbf{x})$, the expected loss relative to a rule based on $\mathbf{x}$ directly is:*

$$\mathbb{E}[\mathcal{L}(\phi \circ \mathcal{A})] \;\geq\; \mathbb{E}[\mathcal{L}(\phi^*)] + \underbrace{I(\mathbf{x};\, \mathbf{x} \mid \mathcal{A}(\mathbf{x}))}_{\textit{information destroyed by } \mathcal{A}}$$

where $I(\cdot;\cdot \mid \cdot)$ is conditional mutual information and ϕ^* is the optimal rule based on $\mathbf{x}$.

This theorem subsumes:

- **Population statistics**: $\mathcal{A}$ = summation over heterogeneous national estimates; micro-level mortality structures destroyed
- **Imperial GDP**: $\mathcal{A}$ = national income accounting; distributional extraction patterns destroyed
- **Phillips Curve**: $\mathcal{A}$ = aggregate π, u; sectoral and expectational heterogeneity destroyed
- **Peer review**: $\mathcal{A}$ = paradigm filter; orthogonal knowledge destroyed per Proposition 6.1

8 Machine Learning Perspective: Overfitting to Paradigm

The gatekeeping pathology has a natural ML analogue: *overfitting*. A model trained exclusively on paradigm-consistent data $\mathcal{D}_\Pi$ achieves low training loss but high generalisation loss on out-of-paradigm evidence $\mathcal{D}_\perp$.

$$\mathcal{L}_{\text{gen}} = \mathcal{L}_{\text{train}} + \underbrace{\Omega(\text{paradigm complexity})}_{\text{regularisation term}} + \underbrace{\text{bias}^2(\hat{\Pi}, \Pi^*)}_{\text{paradigm misspecification}} \tag{6}$$

The bias–variance decomposition (6) shows that reducing training loss by tightening paradigm conformity (increasing Ω) necessarily inflates the bias term if the true data-generating process Π^* differs from the enforced paradigm $\hat{\Pi}$.

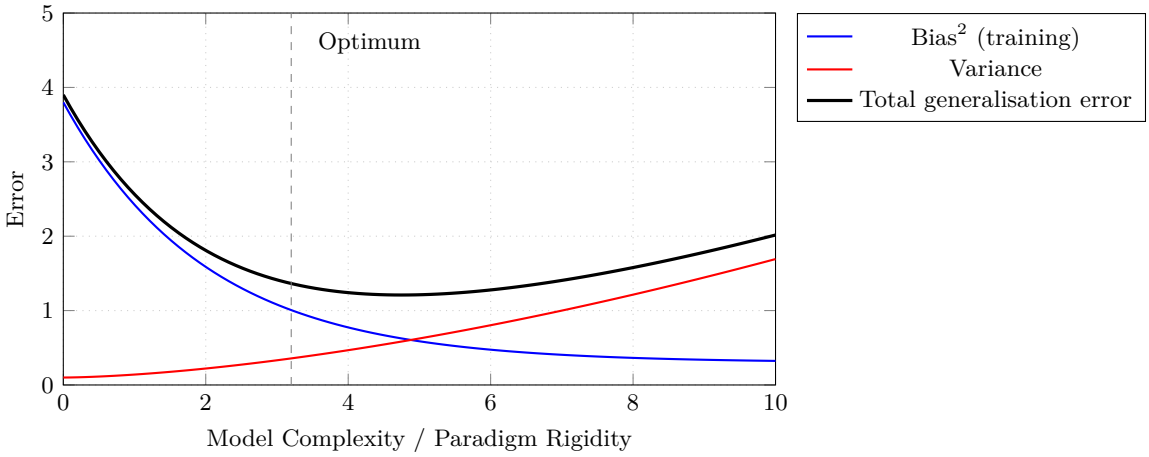

Figure 4: Bias–variance tradeoff applied to paradigm rigidity. Excessive paradigm conformity reduces apparent training error but inflates generalisation error through paradigm misspecification.

9 Conclusion

We have shown that a single epistemological maxim—that statistics constitute the most seductive form of deception—generates a coherent and mathematically tractable research programme spanning demography, imperial history, macroeconomics, and the sociology of knowledge.

The key findings are:

1. Global population estimates carry irreducible uncertainty of $\mathcal{O}(10^8)$ but are triangulated to exclude catastrophic undercounts (Section 2).

2. The Winner's Curse (Theorem 3.1) predicts imperial overextension and post-hegemonic relative decline, consistent with British and Roman trajectories and inconsistent with post-defeat German and Austrian recoveries (Section 3).

3. The empirical Phillips Curve collapses precisely as Goodhart's Law and Austrian price theory predict, vindicating methodological scepticism of aggregate statistical relationships (Section 4, Theorem 4.1).

4. Keynesian short-termism embeds an intergenerational ethical violation: the long run always arrives, populated by persons who bear its costs (Section 5).

5. Institutional peer review is an information-destroying aggregation operator (Proposition 6.1, Theorem 7.1) that systematically suppresses paradigm-orthogonal knowledge (Section 6).

6. The Viennese café 1880–1938 constitutes historical evidence that unregulated epistemic markets outperform gatekept systems by orders of magnitude in output quality (Figure 3).

The unified Aggregation–Distortion Theorem (Theorem 7.1) subsumes all these phenomena as instances of information loss under centralised aggregation—whether statistical, political, or epistemic.

The PhD, as a licence to utter what the institutional paradigm approves, is therefore itself a subject of the theorem it would be required to prove.

References

[1] Disraeli, B. (attributed, c. 1870). Quoted in Twain, M. (1906), *Autobiography*. New York: Harper & Brothers.

[2] Phillips, A.W. (1958). The relation between unemployment and the rate of change of money wage rates in the United Kingdom, 1861–1957. *Economica*, 25(100), 283–299.

[3] Friedman, M. (1968). The role of monetary policy. *American Economic Review*, 58(1), 1–17.

[4] Phelps, E.S. (1968). Money-wage dynamics and labor-market equilibrium. *Journal of Political Economy*, 76(4), 678–711.

[5] Hayek, F.A. (1945). The use of knowledge in society. *American Economic Review*, 35(4), 519–530.

[6] Hayek, F.A. (1974). *The Pretence of Knowledge*. Nobel Memorial Lecture, Stockholm.

[7] von Mises, L. (1949). *Human Action: A Treatise on Economics*. New Haven: Yale University Press.

[8] Keynes, J.M. (1923). *A Tract on Monetary Reform*. London: Macmillan.

[9] Kuhn, T.S. (1962). *The Structure of Scientific Revolutions*. Chicago: University of Chicago Press.

[10] Goodhart, C.A.E. (1975). Problems of monetary management: The U.K. experience. *Papers in Monetary Economics*, Reserve Bank of Australia.

[11] Thaler, R.H. (1988). Anomalies: The winner's curse. *Journal of Economic Perspectives*, 2(1), 191–202.

[12] Capen, E., Clapp, R., & Campbell, W. (1971). Competitive bidding in high-risk situations. *Journal of Petroleum Technology*, 23(6), 641–653.

[13] Ferguson, N. (2003). *Empire: The Rise and Demise of the British World Order*. New York: Basic Books.

[14] Jászi, O. (1929). *The Dissolution of the Habsburg Monarchy*. Chicago: University of Chicago Press.

[15] Feyerabend, P. (1975). *Against Method*. London: New Left Books.

[16] Popper, K.R. (1934). *Logik der Forschung*. Vienna: Julius Springer. (English translation: *The Logic of Scientific Discovery*, London: Hutchinson, 1959.)

[17] Taleb, N.N. (2007). *The Black Swan: The Impact of the Highly Improbable*. New York: Random House.

Glossary

Aggregation–Distortion Theorem The formal result (Theorem 7.1) showing that any aggregation operator destroys micro-level information, increasing expected policy or inference loss relative to rules based on disaggregated data.

Austrian Business Cycle Theory The Misesian–Hayekian account of boom–bust cycles as endogenous consequences of central-bank credit expansion below the natural interest rate, inducing malinvestment that must subsequently be liquidated.

Bias–Variance Tradeoff In statistical learning, the decomposition of generalisation error into irreducible noise, squared bias (systematic misspecification), and variance (sensitivity to training data). Applied here to paradigm rigidity.

NAIRU Non-Accelerating Inflation Rate of Unemployment: the unemployment rate u^* at which inflation is stable; the vertical long-run Phillips Curve passes through this point.

Epistemic Market A decentralised system of knowledge production in which ideas compete on evidential merit without prior gatekeeping. The Viennese café circa 1880–1938 is the canonical historical instance.

Gatekept Knowledge System An institutional arrangement in which centralised authorities (editors, reviewers, grant committees) filter knowledge claims against a prevailing paradigm, as formalised in Proposition 6.1.

Goodhart's Law Any statistical regularity that becomes a policy target ceases to be a reliable regularity, due to endogenous behavioural adaptation.

Intergenerational Discount Rate The rate at which current decision-makers discount the welfare of future generations; the mathematical underpinning of Keynesian short-termism, shown in equation (4).

Malinvestment Austrian term for investment induced by artificially low interest rates that would not be undertaken at the natural rate; the microeconomic content of the boom phase of the Austrian business cycle (equation (5)).

Phillips Curve The empirical and theoretical relationship between unemployment and inflation; stable in 1960s data, collapsed under 1970s stagflation, and near-flat post-2010. See Section 4.

Praxeology Ludwig von Mises's term for the aprioristic science of human action, from which Austrian economics derives its propositions through logical deduction rather than econometric estimation.

Winner's Curse The systematic tendency of auction winners to overpay, because winning is evidence of having submitted the highest—and therefore most overestimating—bid. Generalised here to imperial competition (Theorem 3.1).

The End